Vector Calculus and Complex Analysis Classwork (2025/4/14)

Name:

Student ID:

Find the three possible values of the cube root of the complex number $\frac{1-i}{1+i}$.

Solution It is more convenient to use the polar representation. The polar forms of $1 - i$ and $1 + i$ are

$$1 - i = \sqrt{2}\left(\cos\frac{-\pi}{4} + i\sin\frac{-\pi}{4}\right) \text{ and } 1 + i = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right).$$

Using eq. (1.2.5b), we obtain

$$\frac{1-i}{1+i} = \frac{\sqrt{2}}{\sqrt{2}}\left[\cos\left(\frac{-\pi}{4} - \frac{\pi}{4}\right) + i\sin\left(\frac{-\pi}{4} - \frac{\pi}{4}\right)\right] = \cos\frac{-\pi}{2} + i\sin\frac{-\pi}{2}.$$

By de Moivre's theorem, the three possible values of the cube root of $\dfrac{1-i}{1+i}$ are

$$\cos\frac{-\pi}{6} + i\sin\frac{-\pi}{6},$$

$$\cos\left(\frac{-\pi}{6} + \frac{2\pi}{3}\right) + i\sin\left(\frac{-\pi}{6} + \frac{2\pi}{3}\right) = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i,$$

and

$$\cos\left(\frac{-\pi}{6} + \frac{4\pi}{3}\right) + i\sin\left(\frac{-\pi}{6} + \frac{4\pi}{3}\right) = \cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}.$$